

Bangladesh University of Engineering and Technology
CSE 329

Marks: 20

Time: 25 Minutes

Name:

Student ID:

There are total 3 questions

1. Answer the following question with proper justification **[3*2 = 6]**
 - a. “*K-means algorithm is sensitive to outliers.*” – Do you agree with this statement? If yes can you propose any solution to overcome the problem?
 - b. “*K-means algorithm is sensitive to initial random assignment.*” – Do you agree with this statement? If yes can you propose any solution to overcome the problem?

2. In your class we have discussed three self-supervised methods namely, *Pseudo labeling*, *Expectation-Maximization approach*, and *Co-training*. Now carefully observe the following tasks and state which method works best for which task. Justify your answer only naming the method will no assure any marks. **[3*3 = 9]**
 - a. **Task 1:** You have a very good image classifier model but it is trained using only 20000 image samples. Suddenly, you stumbled upon a dataset with 1M image samples but none of them are labeled. You want to use that dataset to improve your current model. Which method you can use?
 - b. **Task 2:** You are working on a funded project and you have a huge dataset with 1M unlabeled samples. But due to unrest across the globe your funding cut off. Now you have only 5000 tk to hire annotators who can label some of the samples. Cost for annotation is 1tk per sample. Which method will work best to label the rest of the data samples?
 - c. **Task 3:** You have dataset of size 100000X500 (100000 rows and 500 columns). You hired an annotator to label all the samples. Unfortunately, the annotator is very lazy and only labeled 1000 samples. Your project submission deadline is in a week. You have no time to annotate all of them. Which method will work best to label the rest of the data samples?

3. Briefly summarize how PCA algorithm reduce dimensionality of a dataset. Explain why we choose the Eigen vectors with **higher Eigen values** as our final Principal components. **[5]**

